

THE
WELL TEMPERED
MICROPROCESSOR

S 3

BEAR MICROCOMPUTER SYSTEMS



Preface

For many years the act of adding a loudspeaker to a digital computer has inspired musical programs. Now that microcomputers have reached the home, there is perhaps a case for sharing these programs amongst a far wider audience.

This software is intended for amusement and to add a light-hearted moment to what can be a very exacting activity of microcomputing.

The minimum equipment needed consists of:- a 6800 based microcomputer using a MIKBUG* Rom with at least 176 decimal bytes of memory and a F.M. Radio!

If you have enjoyed these programs and write some more the "Bear" will always be interested to hear from you.

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* MIKBUG is a registered trademark of Motorola Ltd.

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THE WELL TEMPERED MICROPROCESSOR

This program will play a tune stored in memory. To run, simply connect a high impedance speaker or earphone to the serial output port of your system. If you do not have such sophisticated equipment, try tuning a VHF radio near the CPU bus lines. You may have to search a bit for the right harmonic. Start the program from location 0030 hex. The program returns to MIKBUG after it has finished.

The tune itself is located from 0068 hex and consists of one note per byte terminated with a zero byte.

The byte format is as follows: The 3 most significant bits represent the length of time of the note or beat. The 5 least significant bits correspond to 31 equally tempered pitches including a silent interval. The actual pitch will be dependent of processor clock rate, but the intervals will remain the same.

There are three Data tapes with the same program preceding them.

Tape 1) ?!!

2) "Happy Birthday to You"

3) Bach's Prelude in C Major.



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HEXADECIMAL CODING FORM

PROGRAM		THE WELL TEMPERED MICROPROCESSOR		1	VERSION	AUTHOR	BEAR	DATE	1-8-77	PAGE	1 OF 3
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS						
0,0,0,0	0,0,F,4,E,6	TABLE	SILENT, C, C#		List of 32 division ratios						
	0,9,C,D,C,2		D, D#, E		00 = silence						
	8,7,A,D,A,3		F, F#, G		01-1F 31 notes on the						
	9,A,9,1,10,9		G#, A, A#		equally tempered scale.						
	8,1,7,A,7,3		B, C, C#		b _s b _s						
	6,C,6,E,6,1		D', D#, E'		0 0 1 0 1 1 0 1 = 						
	5,B,5,6,5,1		F', F#, G'		length note						
	4,D,4,B,4,4		G#, A', A'#		Format of data words						
	4,0,13,0,13,9		B', C", C"#		to be stored from (SCORE+1)						
	3,6,13,3,13,0		D", D"#, E"								
	2,E,12,B,1		F", F#",								
0,0,12,0	1,0,10,0,1	LENGG			{ List of note durations.						
	2,0,10,0,1										
	3,0,10,0,1				TEMPO: Multivibrato or						
	4,0,10,0,1		P		Cristallo controllato						
	5,0,10,0,1				(accordingly)						
	6,0,10,0,1										
	7,0,10,0,1										
	8,0,10,0,1		O								
0,0,3,0	8,E,0,0,6,A	BEGIN	LDS # SCORE		Start here						
0,0,13,3	8,6,7,3,1	NEW	LDAA # 73								
	8,7,0,0,6,2		STAA TIC		Initialise PIA operation						
	3,2,1,1,1		PULA		Get coded note						
	1,6,1,1,1		TAB								
	2,6,10,3,1		BNE OK		Check its not zero						
	7,E,E,0,0,0		JMP \$ E000		twas zero, off to MIKBUG, end tune.						
0,0,13,F	C,4,11,F,1	OK	ANDB # 1F		Mask note pitch,						
	F,7,10,0,15,3		STAB CNT+1		put it where its needed.						
	4,4,1,1,1		LSRA		{ Right justify note length						
	4,4,1,1,1		LSRA								
	4,4,1,1,1		LSRA								
	4,4,1,1,1		LSRA								



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ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS		
	B,4,0,E,1		ANDA # \$ 0E	Form into number (0-E) even.		
	B,8,2,0,1		ADDA # \$ 20	Add displacement.		
	B,7,0,0,5,C		STAA GTMLN+1	Put it where its needed.		
	C,E,0,0,10,0		LDX # TABLE	X contains beginning of delay table.		
0,0,5,2	E,6,0,0,1	GTNT	LDAB 0,X	B contains delay for note time.		
	2,6,0,5,1		BNE GTMLN	Skip if nonzero (silent note)		
	B,6,7,F,1		LDAA # \$ 7F			
	B,7,0,0,6,2		STAA TIC	Change PTA op to clear.		
0,0,5,8	F,F,0,0,1	GTMLN	LDX 0,X	X loaded with note duration.		
	1,7,1,1,1		TBA			
0,0,5,F	5,A,1,1,1	LOOP	DECB	Note generating loop.		
	2,6,0,4,1		BNE CONT			
	1,6,1,1,1		TAB			
0,0,6,2	7,3,8,0,0,4	TIC	COM \$ 8004	Flip-flop PTA		
0,0,6,5	0,9,1,1,1	CONT	DEX			
	2,6,1,F,6,1		BNE LOOP			
0,0,6,8	2,0,1,C,9,1		BRA NEW	End of program		
0,0,6,A	0,0,1,1,1	SCORE		Stack DATA, donot use this byte.		
0,0,6,B	6,8,1,A,D,7,1	MUSIC		Start of data. End with 00 byte.		
	4,F,5,1,4,F					
	A,D,7,1,A,F					
	7,1,A,D,5,1					
	4,F,5,1,4,F					
	A,D,2,0,6,8					
	A,D,7,1,4,F					
	5,1,4,F,A,D			to end of memory		
	7,1,A,F,7,1					
	A,D,5,1,4,F					
	5,1,4,F,A,D					
	9,4,8,6,7,4					
	8,6,7,4,4,F					
	5,1,5,4,8,1					
	2,F,2,0,8,F					

HEXADECIMAL CODING FORM

[illegible]

THE WELL TEMPERED MICROPROCESSOR

?? !!

S113000000F4E6D9CDC2B7ADA39A9189817A736C15
S113001066615856514D4844403D393633302E2B95
S1130020100020003000400050006000700080008C
S11300308E006A8673B70062321626037EE0D0C44F
S11300401FF7005344444444840E8B20B7005CCE15
S11300500000E60D2605867FB70062EE26175A26B5
S113006004167380040926F620C90068AD714F5147
S11300704FAD71AF71AD514F514FAD2068AD714F60
S1130080514FAD71AF71AD514F514FAD94B674B680
S1130090744F5154B12F208F4DAF4D484A4DA81085
S11300A048AA6DA81048AA6DA871EFEF206D00FC56
S105A0480030E2

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HAPPY BIRTHDAY TO YOU

S113000000F4E6D9CDC2B7ADA39A9189817A736C15
S113001066615B56514D4844403D393633302E2B92
S113002006000C00120018001E00230029002F00F7
S11300308E006A8673B70062321626037EE0D0C44F
S11300401FF7005344444444840E8820B7005CCE18
S11300500000E6152605867FB70062EE28175A26AB
S113006004167380040926F620C9009030205092AB
S1130070929090959594949494903020509292906C
S113008090979795959595903020509C9C9999952B
S113009095949492929A3A205A999995959797950E
S11300A0959595809595903020509292909090904F
S10800B09494959500F5
S105A0480030E2

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**THE WELL TEMPERED MICROPROCESSOR
BACH PRELUDE IN C MAJOR**

S113000000F4E6D9CDC2B7ADA39A9189817A736C15
S113001066615B56514D4844403D393633302E2B92
S1130020100020004000400050006000700080007C
S11300308E006A8673B70062321626037EE0D0C44F
S11300401FF7005344444444840E8820B7005CCE18
S11300500000E6092605867FB70062EE24175A26BB
S113006004167380040926F620C900494D505559D9
S1130070505559494B52575A52575A484B50575A50
S113008050575A494D505559505559494D52595E3A
S113009052595E494B4F52574F5257484B50575C39
S11300A050575C48494D50554D505546494D505553
S11300B04D505542464B4F554B4F5544484B505469
S11300C04B505444474D50564D505642464B525750
S11300D04B525742454B4E544B4E54414449505554
S11300E0495055414E52555A52555A464E52555AF8
S11300F052555A444B50545A50545A494D505559DC
S11301005055594950535559535559424E525559C2
S11301105255594349525558525558454E535557BF
S1130120535557444E505457505457444D505559B5
S1130130505559444B50555A50555A444B50545AA3
S113014050545A444C52555B52555B444D50555C87
S113015050555C444B50555A50555A444B50545A80
S113016050545A4449505359505359494B4E52577D
S1130170524E524E4B4E4B484B4884484B4E4B4E7E
S113018057434B3E8D404B4E4D4B49494949490088
S105A0480030E2

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